

A Major Project on
AUTOMATED DETECTION OF SPINAL LESIONS FROM
CT SCANS VIA DEEP TRANSFER LEARNING

A Project Report submitted to
Jawaharlal Nehru Technological University
in partial fulfillment of the requirements for the award of Degree of

Bachelor of Technology
in
COMPUTER SCIENCE AND ENGINEERING

Submitted by

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2025-2026

CERTIFICATE

This is to certify that the Major project entitled “**AUTOMATED DETECTION OF SPINAL LESIONS FROM CT SCANS VIA DEEP TRANSFER LEARNING**” is being submitted by **BADRI ANUSHA**, bearing Roll No.22WJ8A05R2, **GORREPATI CHARANDEEP**, bearing Roll No.22WJ8A05T6, **GUNDU VAMSHI**, bearing Roll No.22WJ8A05U1, **SHAIK NABI**, bearing Roll No.22WJ8A05P8, in partial fulfilment for the award of the Degree of Bachelor of Technology in **Computer Science And Engineering** to the Jawaharlal Nehru Technological University is a record of Bonafide work carried out by them under my guidance and supervision.

The results embodied in this Major project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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PROJECT COMPLETION CERTIFICATE

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The training was conducted on AI BASED PYTHON Technology for the completion of the Project titled “AUTOMATED DETECTION OF SPINAL LESIONS FROM CT SCANS VIA DEEP TRANSFER LEARNING” in JULY 2026. The Project has been completed in all aspects.



DECLARATION OF STUDENT

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I confirm the following:

1. The project was undertaken by us under the supervision of our guide, **Mrs. S. Geetha**, from the selection of the topic to the completion of the final report.
2. I have ensured that the results presented in the report are accurate and based on our original work.
3. Each member of the team has contributed significantly and appropriately to the project work.
4. The project report has been prepared with diligence, ensuring clarity, accuracy, and adherence to academic standards.

we further declare that this report has not been submitted, in part or full, to any other institution or university for the award of any degree or diploma.

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As the guide, I confirm the following:

1. I have overseen the entire project process, from the selection of the project title to the submission of the final report.
2. I have reviewed and certified the accuracy and relevance of the results presented in the report.
3. The contribution of each student have been appropriately recognized and assessed.
4. The project report has been prepared under my supervision, ensuring adherence to high standards of quality, clarity, and structure.

we further certify that this project report has not been previously submitted in part or full for the award of any degree or diploma by any institution or university.

Name of the Guide: Mrs. S. Geetha

Signature of the Guide

Date:

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we would like to express our sincere gratitude to our internal guide, **Mrs. S. Geetha, Assistant Professor of CSE, GNITC**, for her valuable guidance, encouragement, and continuous support throughout the duration of this project.

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ABSTRACT

Automated detection of spinal lesions from Computed Tomographies (CTs) can significantly enhance diagnostic efficiency and accuracy. This project implements a Computer-Aided Detection (CADe) system for spinal lesion analysis using the Xception model. The system is equipped with an intuitive web-based interface developed using the Flask framework, allowing physicians to seamlessly integrate it into their diagnostic workflow. The CAdE system processes input CT scans by extracting vertebral regions of interest, converting them into 2D slices, and performing preprocessing to ensure optimal input quality for the Xception model. The model, fine-tuned with Transfer Learning, classifies vertebrae as either healthy conditions such as metastases, primary tumors, or sclerotic lesions. The training and testing datasets were created from CT scans of patients Records. Data augmentation techniques were applied to expand the dataset and improve model generalization. The Xception model achieved a high accuracy of and a recall of 92.99%, demonstrating its effectiveness in spinal lesion detection. This system aims to provide a reliable and efficient tool to assist medical professionals in spinal lesion diagnosis, enhancing clinical decision-making and patient outcomes.

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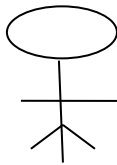
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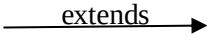
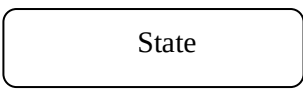
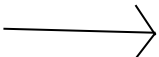
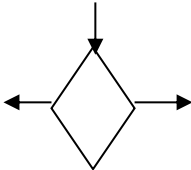
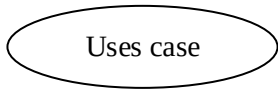
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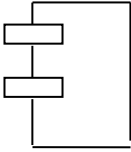
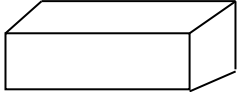
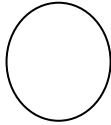
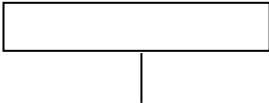
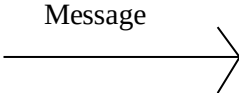
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LIST OF SYMBOLS

S.NO	NOTATION NAME	NOTATION	DESCRIPTION
1.	Class	<i>+ public</i> <i>-private</i> <i># protected</i> <i>Class Name</i> <i>-attribute</i> <i>-attribute</i> <i>+operation</i> <i>+operation</i> <i>+operation</i>	Represents a collection of similar entities grouped together.
2.	Association	Class A NAME Class B Class A — Class B	Associations represents static relationships between classes. Roles represents the way the two classes see each other.
3.	Actor		It aggregates several classes into a single classes.
4.	Aggregation	Class A ↑ Class B Class A ↑ Class B	Interaction between the system and external environment

5.	Relation (uses)	uses	Used for additional process communication.
6.	Relation (extends)		Extends relationship is used when one use case is similar to another use case but does a bit more.
7.	Communication		Communication between various use cases.
8.	State		State of the processes.
9.	Initial State		Initial state of the object
10.	Final state		Final state of the object
11.	Control flow		Represents various control flow between the states.
12.	Decision box		Represents decision making process from a constraint
13.	Use case		Interact ion between the system and external environment.

14.	Component		Represents physical modules which are a collection of components.
15.	Node		Represents physical modules which are a collection of components.
16.	Data Process/State		A circle in DFD represents a state or process which has been triggered due to some event or action.
17.	External entity		Represents external entities such as keyboard, sensors, etc.
18.	Transition		Represents communication that occurs between processes.
19.	Object Lifeline		Represents the vertical dimensions that the object communications.
20.	Message		Represents the message exchanged.

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Computer-Aided Detection (CADe) techniques for the analysis of medical images have grown significantly in importance and sophistication. These tools are particularly critical for extensive exams, such as Computed Tomographies (CTs), which cover large sections of the patient's body and demand meticulous attention from medical professionals. Recently, there has been an increasing focus on automated diagnostic tools powered by Artificial Intelligence (AI). Among these, advanced deep learning models have proven highly effective for analyzing CT, MRI, ultrasound, and radiographic images, leading to state-of-the-art performance in various medical tasks, including spinal pathology detection and classification. Spinal metastases, a type of tumor that originates in other organs and spreads to the spine, are a significant diagnostic challenge. It is estimated that 50% to 70% of cancer patients develop bone metastases, with the spine being one of the most commonly affected regions. Diagnosing spinal tumors using 3D imaging such as CT or MRI is particularly challenging due to the intricate structure of the vertebrae and the sheer number of slices requiring analysis. This highlights the urgent need for a fully automated detection system that can accurately and efficiently identify pathological features in the vertebral body. In this project, we present an automated CADe system utilizing the Xception model, a state-of-the-art deep learning architecture optimized for image classification tasks. Leveraging Transfer Learning, the Xception model is fine-tuned to detect abnormalities in vertebrae, including metastases, primary tumors, and other lesions. The system preprocesses CT scans, extracts vertebral slices, and analyzes them for potential anomalies. To enhance usability, the CADe system incorporates a web-based graphical interface developed using the Flask framework, allowing seamless integration into clinical workflows. This Xception-based solution demonstrates high diagnostic accuracy and provides a reliable, efficient tool to support medical professionals in spinal lesion detection, ultimately improving patient outcomes. Deep learning methodologies have shown promising results in cancer detection, segmentation, and classification, see, e.g., the recent survey [3].

Examples of the different cases of study to which deep learning has been applied include the detection, segmentation, and classification of breast cancer [4], lung cancer [5], brain tumors [6], [7], skin lesion [8], liver tumor [9], [10], and musculoskeletal diseases [11]. Studies about viral and bacterial pneumonia detection from X-rays were made [12]. Latest research focuses on COVID-19 detection from thoracic CT scans [13], [14], [15]. These works consider different imaging problems, in which the inputs come from CT scans [5], [9], [10], [13], [14], [15], X-Rays [12], MRI [6], [7], or RGB images [8]. These methods use open-source networks like, e.g., UNet [16] and SegNet [17] suitably modified. When dealing with small datasets like in medical applications, training can be a difficult task. For this reason, one of the latest approaches is Transfer Learning, i.e. the practice of using state-of-the-art networks that have already been trained from huge datasets and then transferring the achieved knowledge to another domain. Inspired by these approaches, in this paper, we consider the development of a deep-learningbased CAdE system for spine lesions in CT scans. Focusing on the spine, a considerable number of studies regard the identification of the vertebral body volume, see, e.g., the work in [18]. In these works, the goal is to identify and label vertebrae, rather than classify eventual diseases. Several works focus on scoliosis and spondylolisthesis detection and classification, see the survey [19], and the recent works [20], [21], [22]. Differently from our work, these approaches do not focus on vertebral lesions. Moreover, papers in [21] and [22] consider X-Ray images, and the work [20] considers ultrasound data. On the other hand, we consider CT scans. As for the segmentation of intervertebral disks, authors in [23] propose a harmonization approach based on GANs. Differently from our approach, they do not consider lesion classification. Moreover, their method is tailored for MRI images. Lumbar diseases are instead considered in [24]. Similarly to our method, authors propose a classification scheme based on VGG networks. However, they mainly focus on MRI images and do not consider vertebral lesions. A grading system for intervertebral disk degeneration is proposed in [25]. Similarly to the work in [24], authors do not consider vertebral lesions, and their approach is tailored for MRI images. As regards spine tumors, the attention is entirely on spine metastases. Several works consider deep learning models for the detection of abnormalities in the spinal column. In [26], authors propose a study of the performances of a Faster R-CNN network on MRI images. In [27] authors exploit Siamese networks to detect spine metastases from MRI images.

Despite the high performance, the slices have to be manually extracted from the MRI sequences. In [28] MRIs are exploited to label vertebrae and detect the presence of tumors. However, only 2D slices from the sagittal plane are considered, making small or non-centered tumors hard to detect. As regards spine metastasis segmentation, [29], [30] use the UNet neural network on MRI images. Differently from these approaches, in this paper, we consider instead an automatic tool to detect spinal lesions from CT scans rather than MRIs. Also, the proposed methodology does not require manual annotations.

As for deep learning methodologies for detection in CT scans, authors in [31] propose an unbalanced UNet for labeling and segmentation of vertebrae, without a particular focus on abnormalities. The work in [32] presents a pipeline based on a region of interest detector and a multi-modal classifier for spinal metastases. Unlike our approach, that study focuses on metastatic epidural spinal cord compression. In [33], authors propose a diagnosis method for benign and malignant vertebral fractures using the ResNet50 network. However, their pipeline relies on sagittal scans and requires human intervention to select the initial region of interest, which differs from our fully automated system. In [34], a deep learning-based tool employing a convolutional network is used for hybrid detection-segmentation of spinal metastases from CT scans. Although effective, this system was tested exclusively on patients with high levels of metastases, without considering healthy controls. In contrast, our approach evaluates both healthy and lesioned vertebrae. The method in [35] focuses on segmenting metastases from manually extracted volumes of interest in CT scans. Our methodology, however, automates this process without requiring manual extraction. Authors in [36] propose a detection system specifically for lytic spinal lesions using a convolutional model, but their approach excludes other lesion types and does not incorporate a Computer-Aided Detection (CADe) system for automated diagnosis. Similarly, [37] uses an iterative random forest-based method for detecting and classifying metastatic lesions in sagittal scans, which, while achieving good sensitivity, resulted in numerous false positives. For segmentation and detection of sclerotic metastases, the work in [38] utilizes a watershed algorithm on CT scans of the thoracolumbar spine.

In [39], a cascade system is presented that incorporates the pipeline from [38] alongside a specialized CNN trained on CIFAR-10. In contrast, our pipeline is built around the Xception model, a state-of-the-art deep learning architecture pre-trained on a large-scale dataset, which provides superior feature extraction and classification performance.

Our fully automated CAdE system ensures accurate and efficient detection of spinal lesions without reliance on external tools or manual processes, making it a comprehensive solution for spinal anomaly detection.

1.2 SCOPE OF THE PROJECT

This project focuses on developing an advanced automated system for detecting spinal lesions from CT scans using the Xception model. The system identifies abnormalities such as metastases, primary tumors, lytic, and sclerotic lesions with high accuracy and efficiency. By leveraging transfer learning, the Xception model enables effective feature extraction and classification, even with limited medical data, while significantly reducing training time. The project includes data augmentation and preprocessing to improve model performance and generalization. A user-friendly web interface, built using the Flask framework, allows medical professionals to upload CT scans and view results seamlessly.

1.3 OBJECTIVE

The primary objective of this project is to develop an automated system for detecting spinal lesions from CT scans using the Xception model, aiming to assist medical professionals in diagnosing abnormalities accurately and efficiently. The specific objectives include:

1. **Accurate Detection of Spinal Lesions**
To identify and classify spinal abnormalities such as metastases, primary tumors, lytic, and sclerotic lesions with high precision and reliability.
2. **Utilization of Transfer Learning**
To leverage the pre-trained Xception model for efficient feature extraction, reducing training time while maintaining high diagnostic accuracy.
3. **Enhanced Data Handling**
To preprocess CT scan images and apply data augmentation techniques to improve the model's generalization and performance on limited medical datasets.
4. **Development of a User-Friendly Interface**
To create a web-based interface using Flask, enabling medical professionals to upload CT images and access results in a seamless and intuitive manner.
5. **Support for Clinical Decision-Making**
To provide a robust tool that aids radiologists and clinicians in making timely and informed decisions, thereby improving patient outcomes.
6. **Scalability and Future Improvements**
To design a system that can be expanded to include additional imaging modalities and advanced deep learning methods for further enhancements in lesion detection and classification.

1.4 EXISTING SYSTEM

The existing systems for detecting spinal lesions in CT scans often rely on traditional Convolutional Neural Networks (CNNs), such as ResNet50 and VGG, which require manual intervention for region selection and preprocessing. These models, while effective to some extent, often struggle with accuracy and generalization across different lesion types and imaging conditions. Additionally, many of these systems lack a user-friendly interface and require significant human involvement, which can lead to inefficiency and diagnostic errors. Existing systems also tend to be tailored to specific lesion types or imaging modalities, limiting their scalability and flexibility.

1.4.1 EXISTING SYSTEM DISADVANTAGES

- Limited Accuracy: Struggles to work well with different lesion types or imaging conditions.
- Not Scalable: Often designed for specific types of lesions or imaging, making them less flexible.
- Basic Preprocessing: Uses simple techniques that may not work well with medical images.
- Limited Lesion Detection: Usually detects only specific lesion types, missing others.

Slower Processing: Takes longer to analyze CT scans, especially with complex images.

1.5 LITERATURE SURVEY

1. Title: The use of deep learning in medical imaging to improve spine care: A scoping review of current literature and clinical applications

Author: Cody C. Wyles MD, Pouria Rouzrokh MD, Annalise Noelle Larson MD.

Year: 2022.

Description: Medical imaging is critical for clinical decision-making and is integral to determining treatment indications and surgical outcomes. Various imaging modalities can be used for accurate detection and diagnosis of spinal pathologies [91]. However, despite following similar diagnostic standards, experienced radiologists can arrive at different diagnoses and measurements with error rates estimated to be 3% to 5% [7]. Much higher discrepancy rates are reported in neuroradiology, with variable reads in up to 21% of imaging studies [8]. Therefore, a key challenge in the diagnosis of spine pathology is improving the workflow to diagnose diseases quickly, automatically, and accurately. Further, medical care costs are increasing [3], and radiologists who train for 5 or more years after medical school represent an expensive and valuable resource. AI tools to augment the performance of radiologists and provide a low-cost tool to prevent errors hold significant promises for improved quality and cost savings. Reliable treatment planning, precise intervention, and accurate therapy are required when treating spine conditions to achieve optimal outcomes. While modern imaging techniques inform perioperative case management [9], poor outcomes are still frequent. Thus, the development of scalable, perioperative automated assistance tools could help address this challenge.

2. Title: Oncologic Applications of Artificial Intelligence and Deep Learning Methods in CT Spine Imaging.

Author: Wilson Ong, Aric Lee, Wei Chuan Tan.

Year: 2023.

Description: In spinal oncology, integrating deep learning with computed tomography (CT) imaging has shown promise in enhancing diagnostic accuracy, treatment planning, and patient outcomes. This systematic review synthesizes evidence on artificial intelligence (AI) applications in CT imaging for spinal tumors. A PRISMA-guided search identified 33 studies: 12 (36.4%) focused on detecting spinal malignancies, 11 (33.3%) on classification, 6 (18.2%) on prognostication, 3 (9.1%) on treatment planning, and 1 (3.0%) on both detection and classification. Of the classification studies, 7 (21.2%) used machine learning to distinguish between benign and malignant lesions, 3 (9.1%) evaluated tumor stage or grade, and 2 (6.1%) employed radiomics for biomarker classification. Prognostic studies included three (9.1%) that predicted complications such as pathological fractures and three (9.1%) that predicted treatment outcomes. AI's potential for improving workflow efficiency, aiding decision-making, and reducing complications is discussed, along with its limitations in generalizability, interpretability, and clinical integration. Future directions for AI in spinal oncology are also explored. In conclusion, while AI technologies in CT imaging are promising, further research is necessary to validate their clinical effectiveness and optimize their integration into routine practice.

3. Title: Deep learning system for automated detection of posterior ligamentous complex injury in patients with thoracolumbar fracture on MRI

Author: Sang Won Jo, Eun Kyung Khil,

Year: 2023.

Description: This study aimed to develop a deep learning (DL) algorithm for automated detection and localization of posterior ligamentous complex (PLC) injury in patients with acute thoracolumbar (TL) fracture on magnetic resonance imaging (MRI) and evaluate its diagnostic performance. In this retrospective multicenter study, using midline sagittal T2-weighted image with fracture ($\pm$ PLC injury), a training dataset and internal and external validation sets of 300, 100, and 100 patients, were constructed with equal numbers of injured and normal PLCs. The DL algorithm was developed through two steps (Attention U-net and Inception-ResNet-V2). We evaluate the diagnostic performance for PLC injury between the DL algorithm and radiologists with different levels of experience. The area under the curves (AUCs) generated by the DL algorithm were 0.928, 0.916 for internal and external validations, and by two radiologists for observer performance test were 0.930, 0.830, respectively. Although no significant difference was found in diagnosing PLC injury between the DL algorithm and radiologists, the DL algorithm exhibited a trend of higher AUC than the radiology trainee. Notably, the radiology trainee's diagnostic performance significantly improved with DL algorithm assistance. Therefore, the DL algorithm exhibited high diagnostic performance in detecting PLC injuries in acute TL fractures.

4.Title: Deep learning-based automated spine fracture type identification with Clinically validated GAN generated CT images

Author: ORCID Icon,Radhika M. PaiORCID Icon,Shyamasunder N. BhatORCID Icon
&Manohara Pai M. M.ORCID Icon,

Year: 2022

Description: Automatic type identification of sub-axial spine fractures is of prime importance for orthopedic to reduce image interpretation time and increase patient care time. But identifying fracture types is challenging due to imbalanced datasets. In this work, CT scan images of fractured spine has been collected from a Tertiary Care hospital and extended Deep Convolutional Generative Adversarial Network (DCGAN) architecture is developed for generating spine fracture images that overcomes the imbalanced dataset problem. These enhanced datasets are clinically evaluated with Two Visual Turing Tests (VTTs): the first test to “identify real and generated images” and second test to determine “type of fractures in the generated images.” The first VTT demonstrates that generated images of fractures are realistic and that even spine surgeons have difficulty in distinguishing them from real. The second VTT demonstrates that fracture lines are clearly visible in the generated images. The VTT results are analyzed using Fleiss Kappa statistical techniques to determine the inter-observer reliability of spine surgeons’ clinical evaluation. The results showed high interobserver agreement for “type identification” in the generated images. The clinically evaluated generated images are provided to the proposed ensemble-based type identification model, which outperformed other models in type identification.

5.Title: CT-Based Automatic Spine Segmentation Using Patch-Based Deep Learning.

Author: Syed Furqan Qadri, Hongxiang Lin, Linlin Shen

Year: 2022.

Description: CT vertebral segmentation plays an essential role in various clinical applications, such as computer-assisted surgical interventions, assessment of spinal abnormalities, and vertebral compression fractures. Automatic CT vertebral segmentation is challenging due to the overlapping shadows of thoracoabdominal structures such as the lungs, bony structures such as the ribs, and other issues such as ambiguous object borders, complicated spine architecture, patient variability, and fluctuations in image contrast. Deep learning is an emerging technique for disease diagnosis in the medical field. This study proposes a patch-based deep learning approach to extract the discriminative features from unlabeled data using a stacked sparse autoencoder (SSAE). 2D slices from a CT volume are divided into overlapping patches fed into the model for training. A random under sampling (RUS)-module is applied to balance the training data by selecting a subset of the majority class. SSAE uses pixel intensities alone to learn high-level features to recognize distinctive features from image patches. Each image is subjected to a sliding window operation to express image patches using autoencoder high-level features, which are then fed into a sigmoid layer to classify whether each patch is a vertebra or not. We validate our approach on three diverse publicly available datasets: VerSe, CSI-Seg, and the Lumbar CT dataset. Our proposed method outperformed other models after configuration optimization by achieving 89.9% in precision, 90.2% in recall, 98.9% in accuracy, 90.4% in F-score, 82.6% in intersection over union (IoU), and 90.2% in Dice coefficient (DC). The results of this study demonstrate that our model's performance consistency using a variety of validation strategies is flexible, fast, and generalizable, making it suited for clinical application.

1.6 PROPOSED SYSTEM

The proposed system uses the advanced Xception model, which leverages separable convolutions to improve feature extraction and overall performance. It eliminates the need for manual region selection and can automatically detect and classify a wide range of spinal lesions, including metastases, tumors, and fractures. The system also includes a user-friendly interface built with Flask, allowing medical professionals to easily upload CT scans and receive diagnostic results. Furthermore, the proposed system incorporates advanced preprocessing techniques, such as data augmentation, to enhance its accuracy and adaptability to varying medical imaging data. Overall, the proposed system offers a more automated, accurate, and scalable solution for spinal lesion detection compared to existing methods.

1.6.1 PROPOSED SYSTEM ADVANTAGES:

- Better Accuracy: The Xception model improves accuracy and handles different lesion types and conditions better.
- Flexible and Scalable: Can detect various types of spinal lesions and work with different imaging modalities.
- Easy to Use: Simple web-based interface that doctors can easily operate without technical expertise.
- Advanced Preprocessing: Uses better techniques like data augmentation for handling medical images effectively.
- Comprehensive Detection: Detects many types of lesions, not just a few, giving a more complete diagnosis.
- Faster Processing: Analyzes CT scans quickly, improving diagnostic speed.

CHAPTER 2

PROJECT DESCRIPTION

2.1 GENERAL:

2.2 METHODOLOGIES

2.2.1 MODULES NAME:

Modules Name:

- Data Collection
- Data Analysis/load Data
- Data Feature Extraction
- U-net Model for Segmentation
- Split Dataset
- Xception Model for Training
- Model Evaluation
- Model Save

2.2.2 MODULES EXPLANATION:

1) **Data Collection:** This step involves gathering ultrasound image data related to histotripsy ablation. The data could include images of both treated (ablated) and untreated (non-ablated) tissue, usually acquired from patients undergoing the histotripsy procedure or from publicly available datasets.

2) **Data Analysis/Load Data:** After collecting the dataset, the next step is analyzing and loading the data into the system. This involves checking the quality of the data, ensuring there are no missing or corrupted images, and formatting them correctly for further processing. In this step, the images are typically resized and normalized for consistency.

3) **Data Feature Extraction:** In this step, the system uses various techniques to extract meaningful features from the ultrasound images. Features such as texture, edges, or regions with high intensity (indicating potential ablation) are identified. This helps the Xception in the next step to learn essential patterns from the images.

4) **U-Net Model for Segmentation:** The U-Net model is a deep learning architecture specifically designed for image segmentation. It consists of an encoder (which captures feature representations of the image) and a decoder (which reconstructs the segmentation output). The U-Net is used to segment the ultrasound images into regions of ablated and non-ablated tissues, providing a pixel-wise classification.

5) **Split Dataset:** To train and evaluate the model, the dataset is divided into three parts: training, validation, and testing datasets. Typically, the dataset is split such that 60% is used for training the model, 20% for validating the model during training (to tune hyperparameters), and the remaining 20% for testing the final model's performance.

6) **Xception Model for Training:** In this step, the Xception model is trained using the segmented images from the U-Net model. The Xception is responsible for learning high-level features from the images and classifying them as ablated or non-ablated tissues. The model uses the images from the training dataset to learn patterns that are then applied to unseen test data.

7) **Model Evaluation:** After training the Xception, the model is evaluated using the test dataset. Metrics such as accuracy, precision, recall, and the Dice similarity coefficient are used to assess how well the model performs. This step helps in determining whether the model has learned to correctly identify ablation areas and if further tuning or adjustments are needed.

8) **Model Save:** Once the model achieves satisfactory performance, it is saved for future use. The trained model can be saved as a file (e.g., .h5 or .pkl) and later loaded to make predictions on new, unseen ultrasound images. Saving the model ensures that it does not have to be retrained every time it needs to be used in a real-time application.

2.3 TECHNIQUE USED OR ALGORITHM USED

2.3.1 EXISTING TECHNIQUE: -

➤ CNN Models

In the existing system, the workflow begins with the acquisition of a CT scan of the patient's spine using a tomography machine. Once the scan is obtained, manual intervention is required as the physician or technician must select the regions of interest (ROI), such as the vertebrae, from the scan. This step is crucial for ensuring the correct focus on the relevant areas of the scan but can be time-consuming and prone to human error. After the ROI is selected, basic preprocessing techniques, such as resizing and normalization, are applied to prepare the data for further analysis. Next, a pre-trained Convolutional Neural Network (CNN) model, like ResNet50 or VGG, is used to extract features from the processed images. The extracted features are then analyzed by the model to classify the vertebrae as healthy or containing lesions, such as metastases or fractures. Following feature extraction and analysis, the model classifies the vertebrae into different categories, such as healthy or various types of lesions. The results are then displayed to the physician, who makes the final diagnosis, often requiring further manual analysis. This workflow involves significant human involvement at multiple stages, making it less efficient and more prone to errors, particularly when handling complex or varied imaging conditions.

2.3.2 PROPOSED TECHNIQUE USED OR ALGORITHM USED:

➤ XCEPTION MODEL

The proposed system operates with a fully automated approach to detect spinal lesions from CT scans, leveraging the Xception model for image classification. The process begins with the acquisition of a CT scan of the patient's spine, which is then processed automatically without the need for manual intervention. Unlike the existing system, the system automatically identifies and extracts the relevant vertebral regions from the CT scan. Once the regions are extracted, advanced preprocessing techniques, including data augmentation, are applied to improve the quality and variability of the data. The preprocessed images are then passed through the Xception model, a deep learning architecture known for its efficient performance in image classification tasks. The Xception model, pre-trained on large datasets, performs feature extraction from the images and classifies the vertebrae into different categories, such as healthy or containing lesions like abnormal and normal. The results of the classification are presented through a user-friendly web-based interface, allowing physicians to quickly review the findings. The system automatically generates a report with the diagnosis, which the physician can use as a support tool in making final decisions. This proposed system offers enhanced accuracy, faster processing, and a comprehensive detection capability, improving upon the existing system by eliminating the need for manual input and improving overall diagnostic efficiency.

CHAPTER 3

REQUIREMENTS ENGINEERING

3.1 GENERAL

We can see from the results that on each database, the error rates are very low due to the discriminatory power of features and the regression capabilities of classifiers. Comparing the highest accuracies (corresponding to the lowest error rates) to those of previous works, our results are very competitive.

3.2 HARDWARE REQUIREMENTS

The hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system. They are used by software engineers as the starting point for the system design. It should what the system do and not how it should be implemented.

- PROCESSOR : DUAL CORE 2 DUOS.
- RAM : 4GB DD RAM
- HARD DISK : 250 GB

3.3 SOFTWARE REQUIREMENTS

The software requirements document is the specification of the system. It should include both a definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks and tracking the teams and tracking the team's progress throughout the development activity.

- Operating System : Windows 7/8/10
- Platform : Spyder3
- Programming Language : Python
- Front End : Spyder3

3.4 FUNCTIONAL REQUIREMENTS

A functional requirement defines a function of a software-system or its component. A function is described as a set of inputs, the behavior, Firstly, the system is the first that achieves the standard notion of semantic security for data confidentiality in attribute-based deduplication systems by resorting to the hybrid cloud architecture.

3.5 NON-FUNCTIONAL REQUIREMENTS

The major non-functional Requirements of the system are as follows

Usability

The system is designed with completely automated process hence there is no or less user intervention.

Reliability

The system is more reliable because of the qualities that are inherited from the chosen platform python. The code built by using python is more reliable.

Performance

This system is developing in the high level languages and using the advanced back-end technologies it will give response to the end user on client system with in very less time.

Supportability

The system is designed to be the cross platform supportable. The system is supported on a wide range of hardware and any software platform, which is built into the system.

Implementation

The system is implemented in web environment using Jupyter notebook software. The server is used as the intelligence server and windows 10 professional is used as the platform.

Interface the user interface is based on Jupyter notebook provides server system.

CHAPTER 4

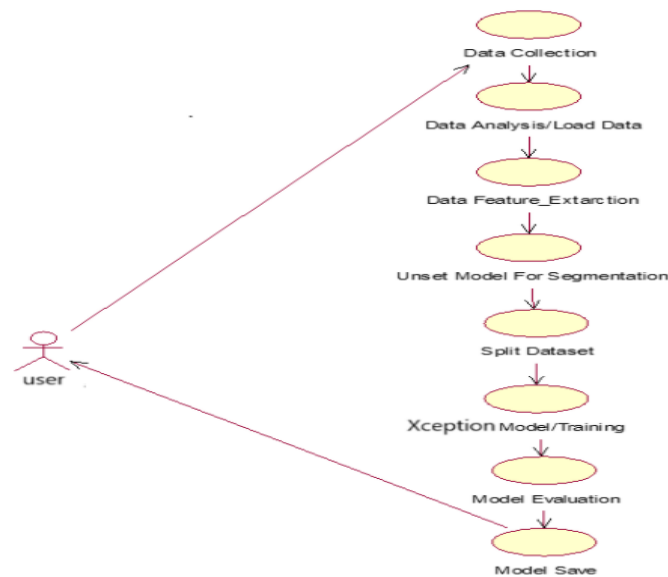
SYSTEM DESIGN

4.1 GENERAL

Design Engineering deals with the various UML [Unified Modelling language] diagrams for the implementation of project. Design is a meaningful engineering representation of a thing that is to be built. Software design is a process through which the requirements are translated into representation of the software. Design is the place where quality is rendered in software engineering.

4.2 UML DIAGRAMS

4.2.1 USE CASE DIAGRAM

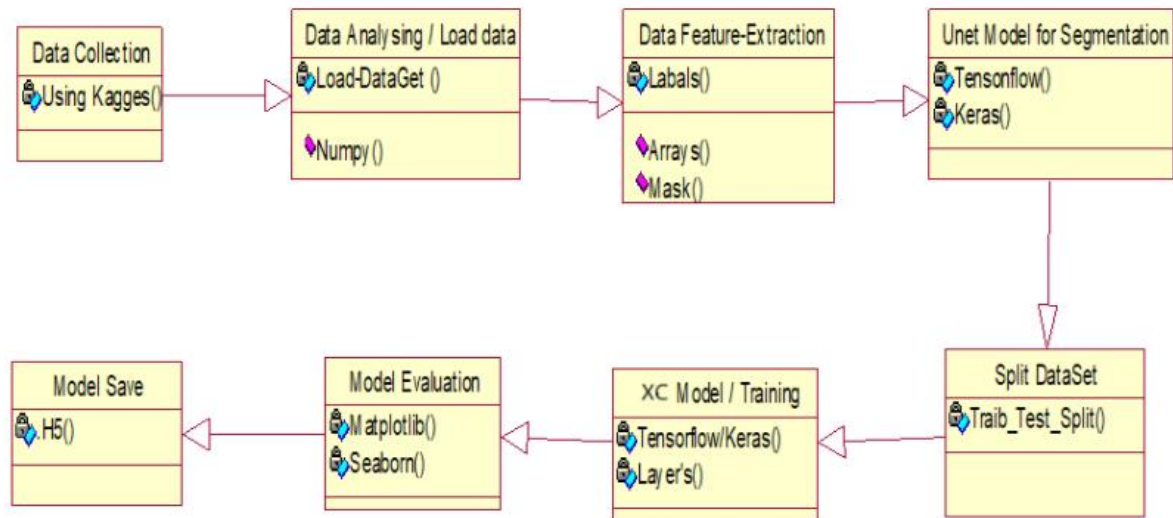


EXPLANATION:

The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted. The above diagram consists of user as actor. Each will play a certain role to achieve the concept.

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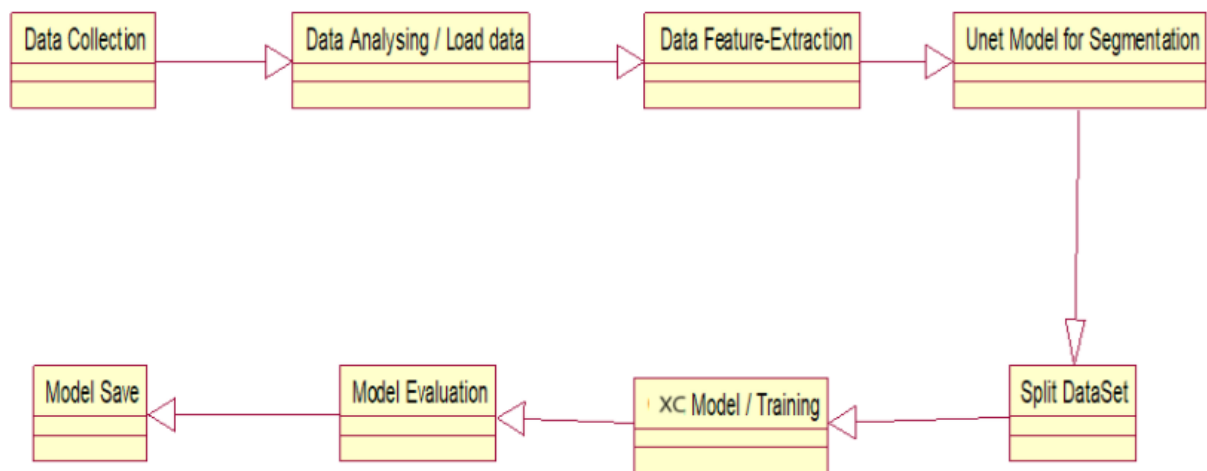
4.2.2 CLASS DIAGRAM



EXPLANATION

In this class diagram represents how the classes with attributes and methods are linked together to perform the verification with security. From the above diagram shown the various classes involved in our project.

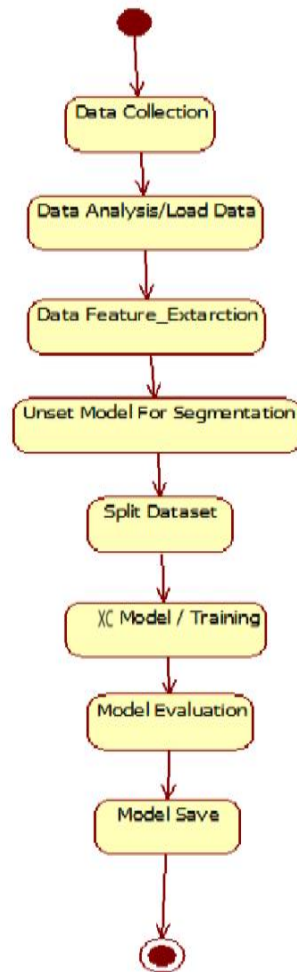
4.2.3 OBJECT DIAGRAM



EXPLANATION:

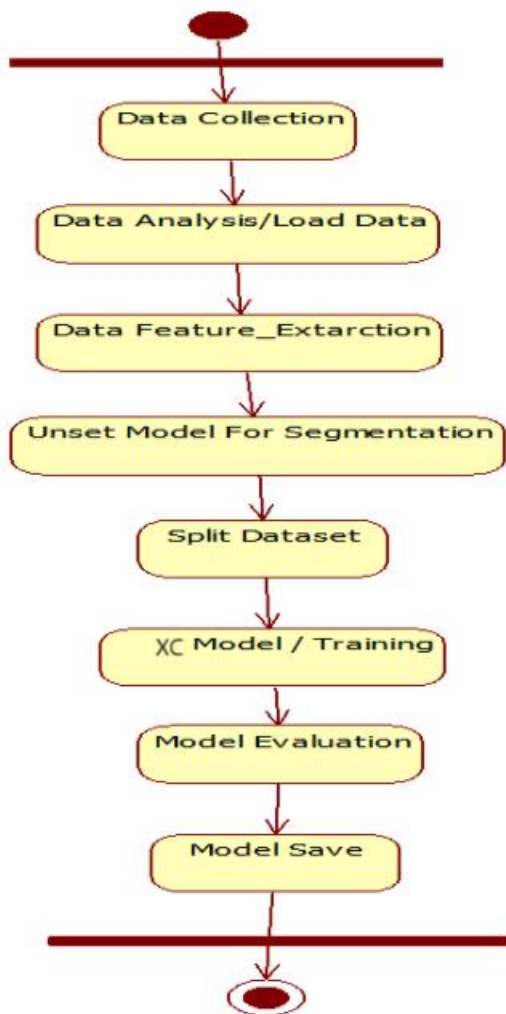
In the above diagram tells about the flow of objects between the classes. It is a diagram that shows a complete or partial view of the structure of a modeled system. In this object diagram represents how the classes with attributes and methods are linked together to perform the verification with security.

4.2.4 STATE DIAGRAM

**EXPLANATION:**

State diagram are a loosely defined diagram to show workflows of stepwise activities and actions, with support for choice, iteration and concurrency. State diagrams require that the system described is composed of a finite number of states; sometimes, this is indeed the case, while at other times this is a reasonable abstraction. Many forms of state diagrams exist, which differ slightly and have different semantics.

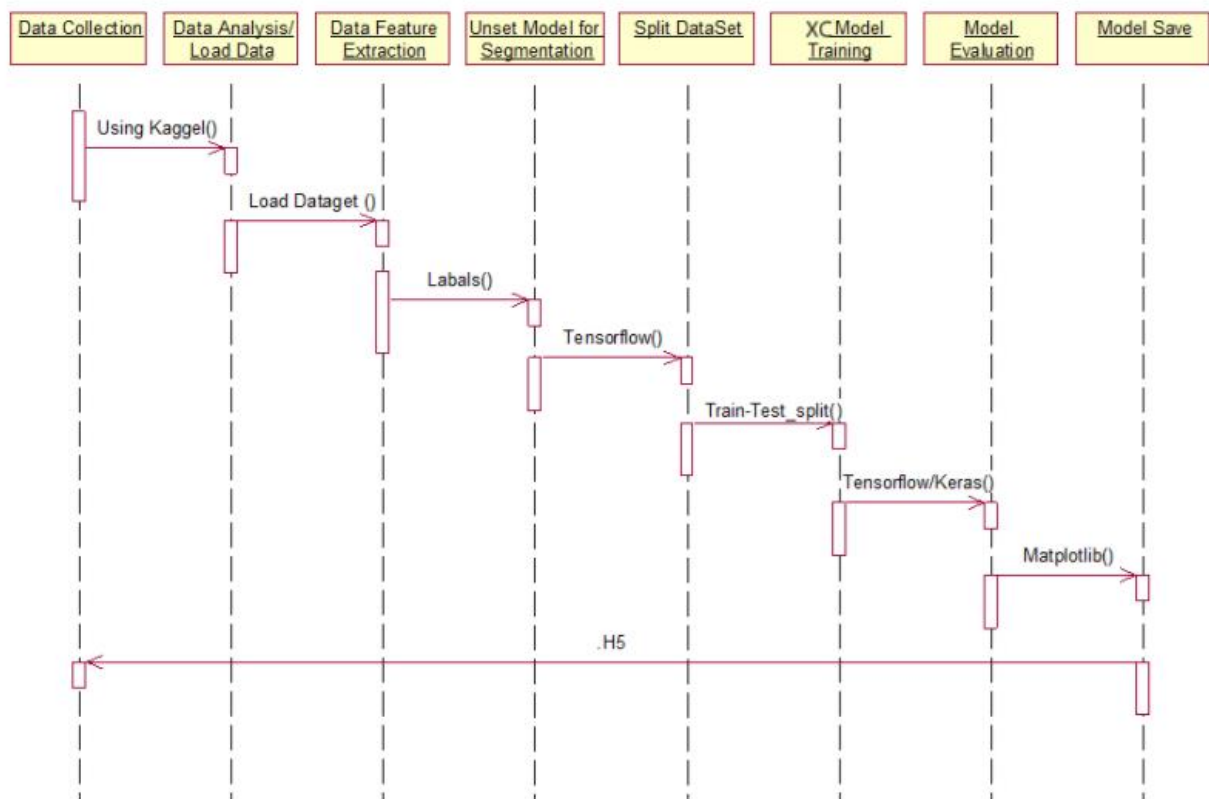
4.2.5 ACTIVITY DIAGRAM



EXPLANATION:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

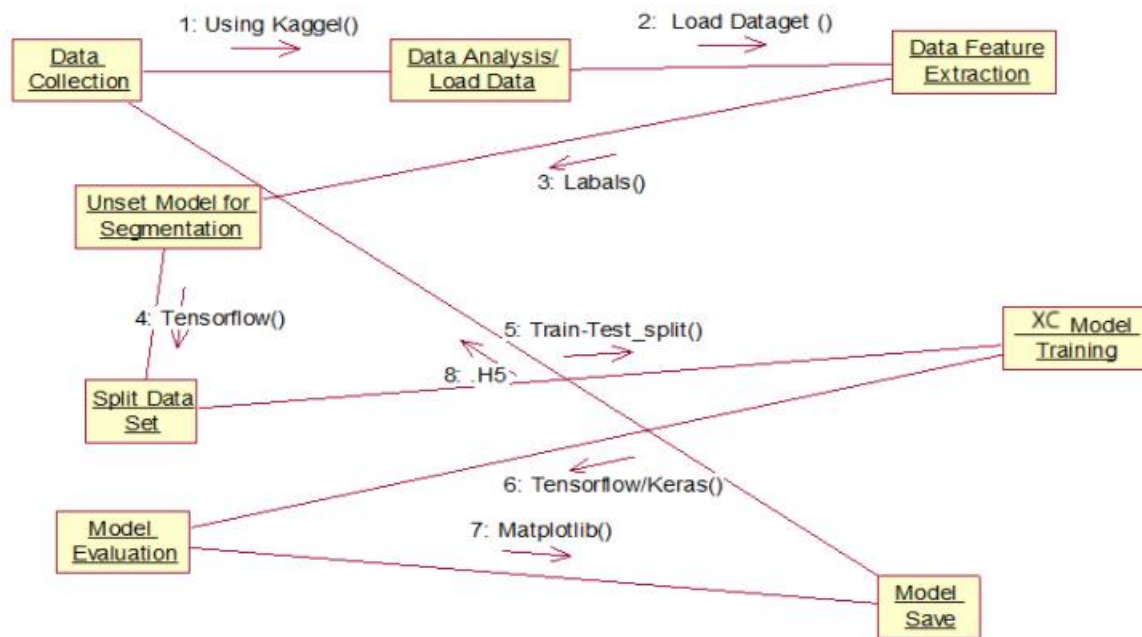
4.2.6 SEQUENCE DIAGRAM



EXPLANATION:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario.

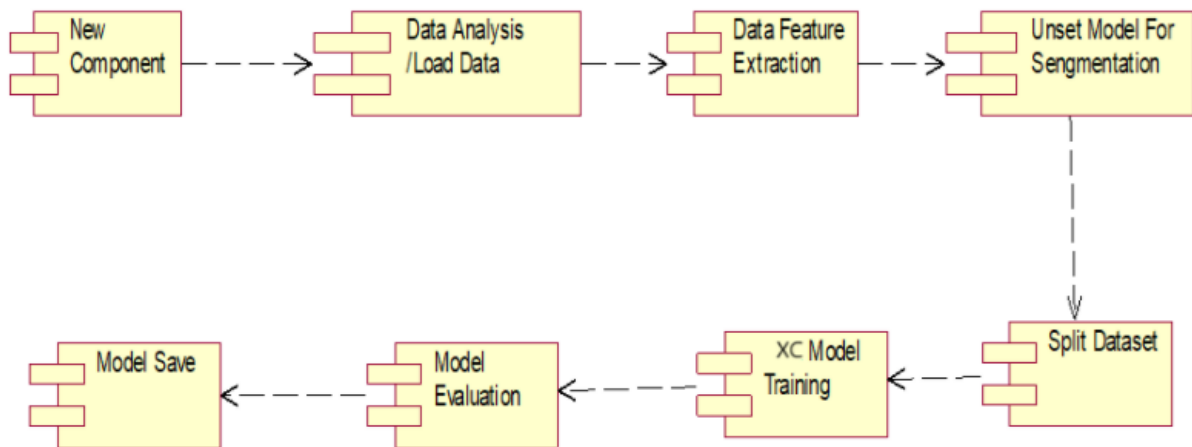
4.2.7 COLLABORATION DIAGRAM



EXPLANATION:

A collaboration diagram, also called a communication diagram or interaction diagram, is an illustration of the relationships and interactions among software objects in the Unified Modeling Language (UML). The concept is more than a decade old although it has been refined as modeling paradigms have evolved.

4.2.8 COMPONENT DIAGRAM

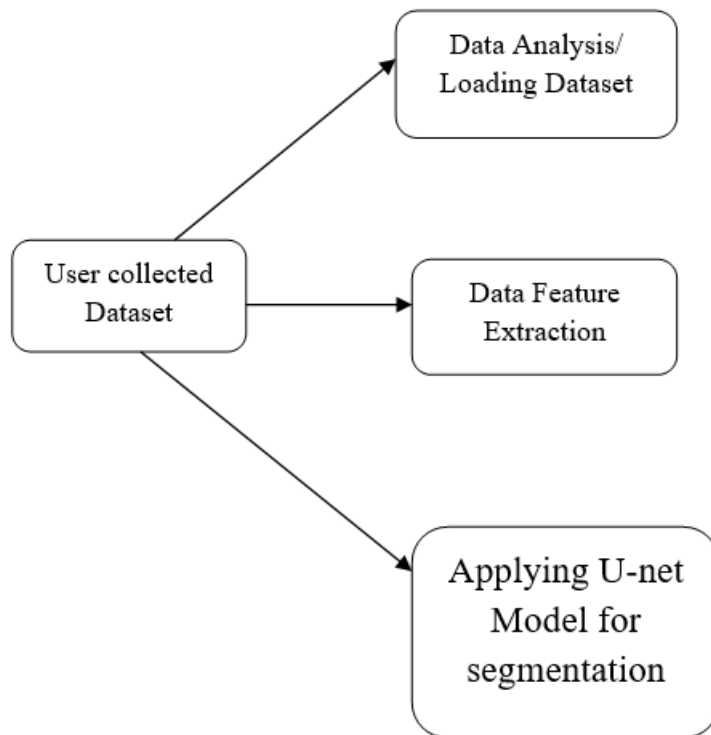


EXPLANATION

In the Unified Modeling Language, a component diagram depicts how components are wired together to form larger components and or software systems. They are used to illustrate the structure of arbitrarily complex systems. User gives main query and it converted into sub queries and sends through data dissemination to data aggregators. Results are to be showed to user by data aggregators. All boxes are components and arrow indicates dependencies.

4.2.9 DATA FLOW DIAGRAM

Level 0



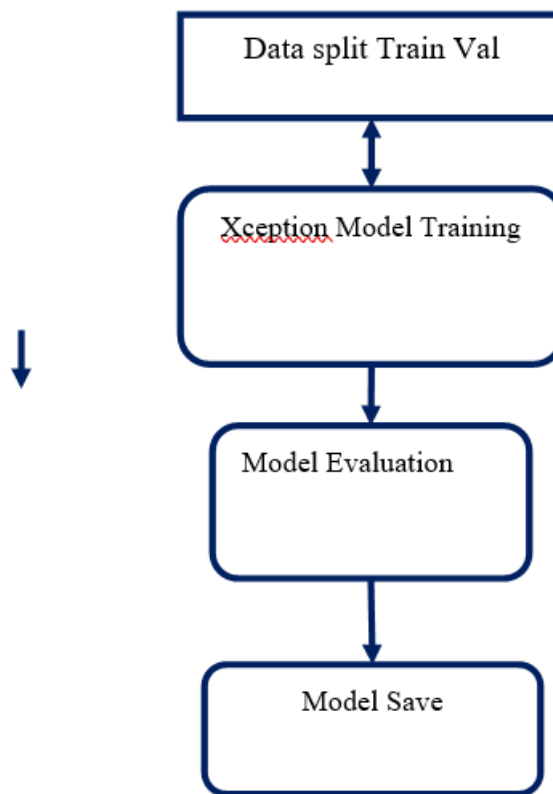
Level 1

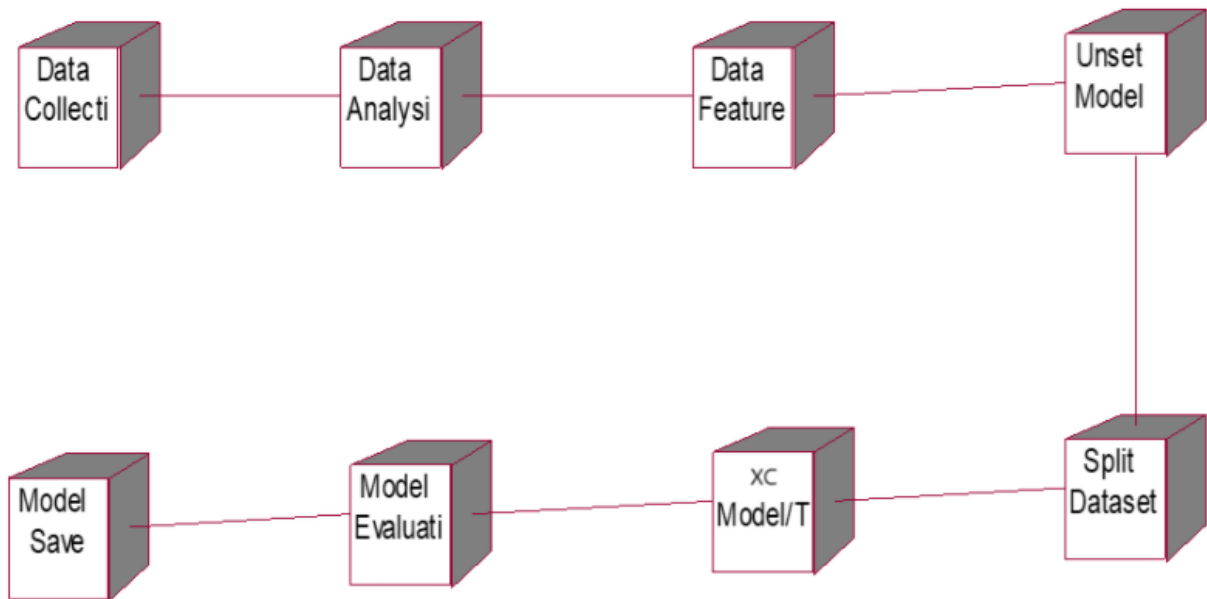
Fig 4.9: Data Flow Diagrams

EXPLANATION:

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. Often they are a preliminary step used to create an overview of the system which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design).

A DFD shows what kinds of data will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of processes, or information about whether processes will operate in sequence or in parallel.

4.2.10 DEPLOYMENT DIAGRAM



EXPLANATION:

Deployment Diagram is a type of diagram that specifies the physical hardware on which the software system will execute. It also determines how the software is deployed on the underlying hardware. It maps software pieces of a system to the device that are going to execute it.

4.2.11 SYSTEM ARCHITECTURE:

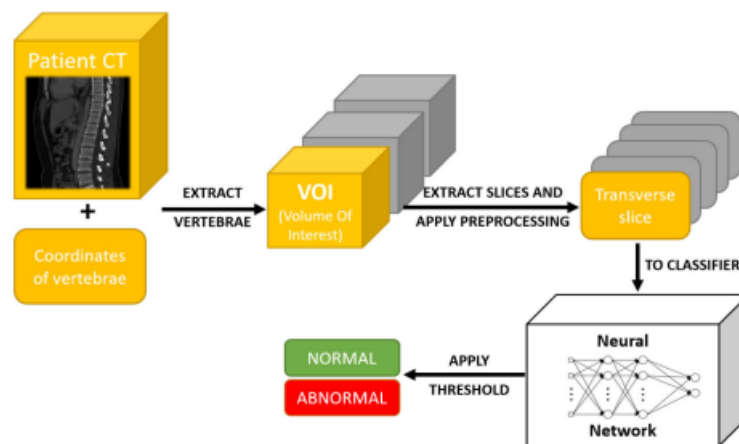


Fig 4.2.11: System Architecture

CHAPTER 5

DEVELOPMENT TOOLS

5.1 Python

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

5.2 History of Python

- Python was developed by Guido van Rossum in the late eighties and early nineties at the National Research Institute for Mathematics and Computer Science in the Netherlands.
- Python is derived from many other languages, including ABC, Modula-3, C, C++, Algol-68, SmallTalk, and Unix shell and other scripting languages.
- Python is copyrighted. Like Perl, Python source code is now available under the GNU General Public License (GPL).
- Python is now maintained by a core development team at the institute, although Guido van Rossum still holds a vital role in directing its progress.

5.3 Importance of Python

- **Python is Interpreted** – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- **Python is Interactive** – You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- **Python is Object-Oriented** – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- **Python is a Beginner's Language** – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

5.4 Features of Python

- **Easy-to-learn** – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- **Easy-to-read** – Python code is more clearly defined and visible to the eyes.
- **Easy-to-maintain** – Python's source code is fairly easy-to-maintain.
- **A broad standard library** – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode** – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable** – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- **Extendable** – You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- **Databases** – Python provides interfaces to all major commercial databases.
- **GUI Programming** – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- **Scalable** – Python provides a better structure and support for large programs than shell scripting.

Apart from the above-mentioned features, Python has a big list of good features, few are listed below –

- It supports functional and structured programming methods as well as OOP.
- It can be used as a scripting language or can be compiled to byte-code for building large applications.
- It provides very high-level dynamic data types and supports dynamic type checking.
- IT supports automatic garbage collection.
- It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.

5.5 Libraries used in python

- **numpy** - mainly useful for its N-dimensional array objects.
- **pandas** - Python data analysis library, including structures such as dataframes.
- **matplotlib** - 2D plotting library producing publication quality figures.
- **scikit-learn** - the machine learning algorithms used for data analysis and data mining tasks.



Figure : NumPy, Pandas, Matplotlib, Scikit-learn

CHAPTER 6

IMPLEMENTATION

6.1 GENERAL

6.2 CODING:

```
import os
from flask import Flask, request, render_template, send_from_directory
from werkzeug.utils import secure_filename
import numpy as np
import cv2
from tensorflow.keras.models import load_model
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
# Initialize Flask App
app = Flask(__name__)
# Configuration
UPLOAD_FOLDER = 'uploads'
RESULT_FOLDER = 'static/results' # Store results inside static folder for direct access
os.makedirs(UPLOAD_FOLDER, exist_ok=True)
os.makedirs(RESULT_FOLDER, exist_ok=True)
app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER
app.config['RESULT_FOLDER'] = RESULT_FOLDER
ALLOWED_EXTENSIONS = {'png', 'jpg', 'jpeg'}
# Load the trained model
model = load_model('segmentation_classification_model.h5')
# Load the label encoder
label_encoder = LabelEncoder()
label_encoder.classes_ = np.load('classes.npy') # Save and load your classes
```

```
# Function to check allowed file extensions
def allowed_file(filename):
    return '.' in filename and filename.rsplit('.', 1)[1].lower() in ALLOWED_EXTENSIONS

# Function to preprocess the image
def preprocess_image(image_path, target_size=(256, 256)):
    img = cv2.imread(image_path)
    img_resized = cv2.resize(img, target_size)
    img_normalized = img_resized / 255.0
    return np.expand_dims(img_normalized, axis=0), img_resized

# Route for homepage
@app.route('/')
def index():
    return render_template('index.html')

# Route for handling image upload and prediction
@app.route('/predict', methods=['POST'])
def predict():
    if 'file' not in request.files:
        return "No file uploaded."
    file = request.files['file']
    if file.filename == "" or not allowed_file(file.filename):
        return "Invalid file type. Please upload a PNG, JPG, or JPEG file."
    # Save uploaded file
    filename = secure_filename(file.filename)
    file_path = os.path.join(app.config['UPLOAD_FOLDER'], filename)
    file.save(file_path)
    # Preprocess the image
    input_image, original_image = preprocess_image(file_path)
    # Make predictions
    prediction = model.predict(input_image)
    predicted_mask = prediction[0][0, :, :, 0] # Segmentation output
    predicted_label = label_encoder.inverse_transform([np.argmax(prediction[1])])[0]
```

```
# Apply a color map to the mask for better visualization
colored_mask = cv2.applyColorMap((predicted_mask * 255).astype(np.uint8),
cv2.COLORMAP_JET)

# Save the results
original_result_path = os.path.join(app.config['RESULT_FOLDER'], 'original_' +
filename)
colored_segmented_result_path = os.path.join(app.config['RESULT_FOLDER'],
'segmented_' + filename)

# Normalize paths to use forward slashes
original_result_path = original_result_path.replace("\\", "/")
colored_segmented_result_path = colored_segmented_result_path.replace("\\", "/")

# Save images
cv2.imwrite(original_result_path, original_image)
cv2.imwrite(colored_segmented_result_path, colored_mask)

# Render results
return render_template('result.html',
                        original_image=original_result_path,
                        segmented_image=colored_segmented_result_path,
                        predicted_label=predicted_label)

# Route to serve images from the 'static' folder
@app.route('/static/results/<filename>')
def result_file(filename):
    return send_from_directory(app.config['RESULT_FOLDER'], filename)

if __name__ == '__main__':
    app.run(debug=True)
```

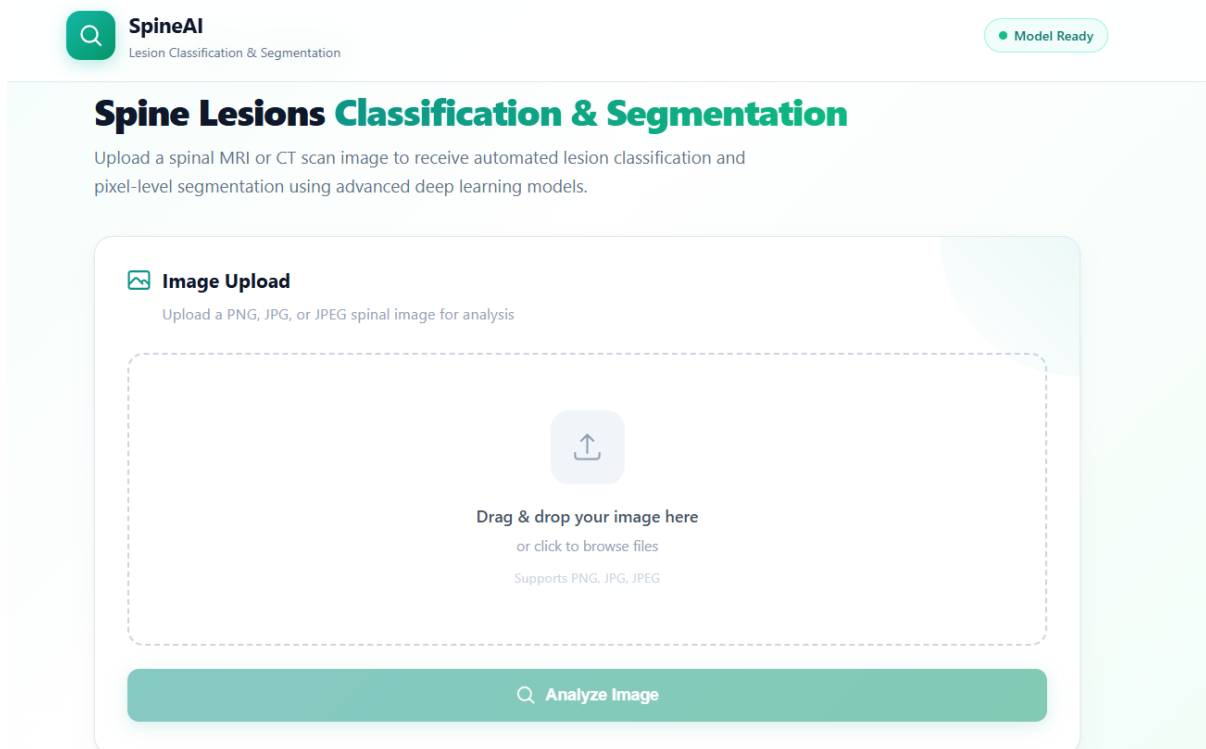
CHAPTER 7

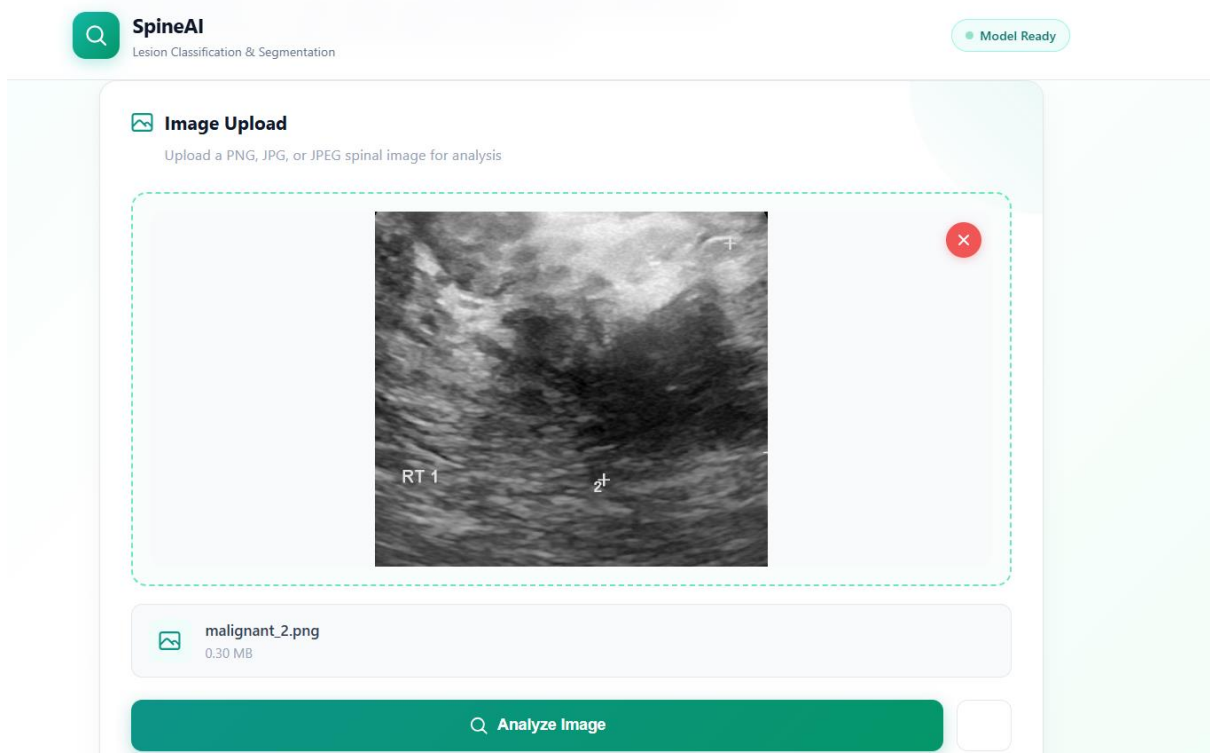
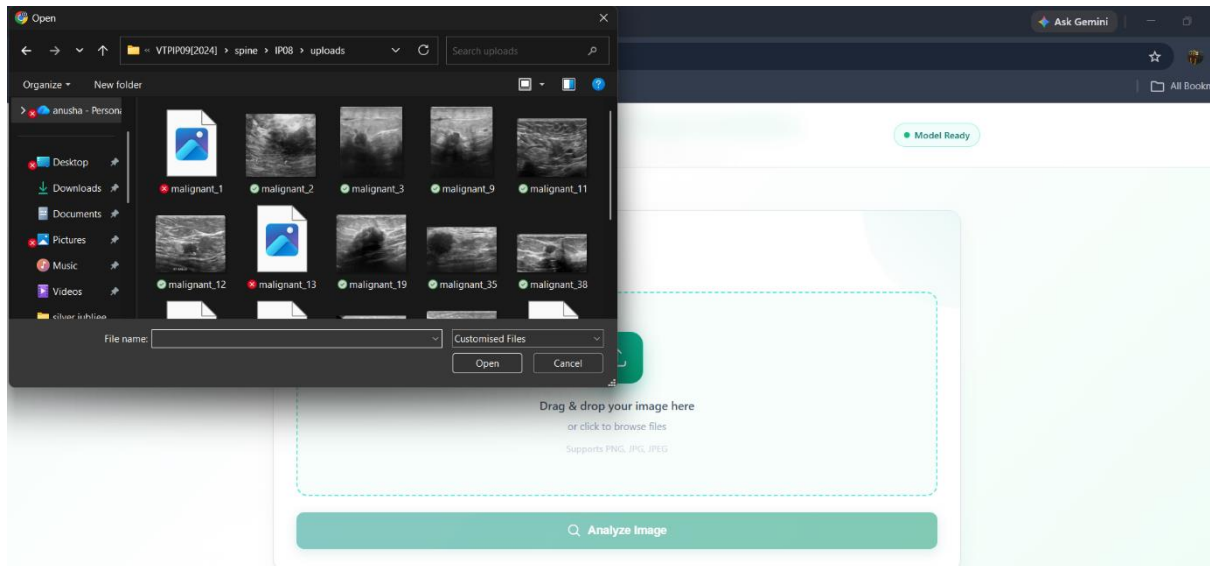
SNAPSHOTS

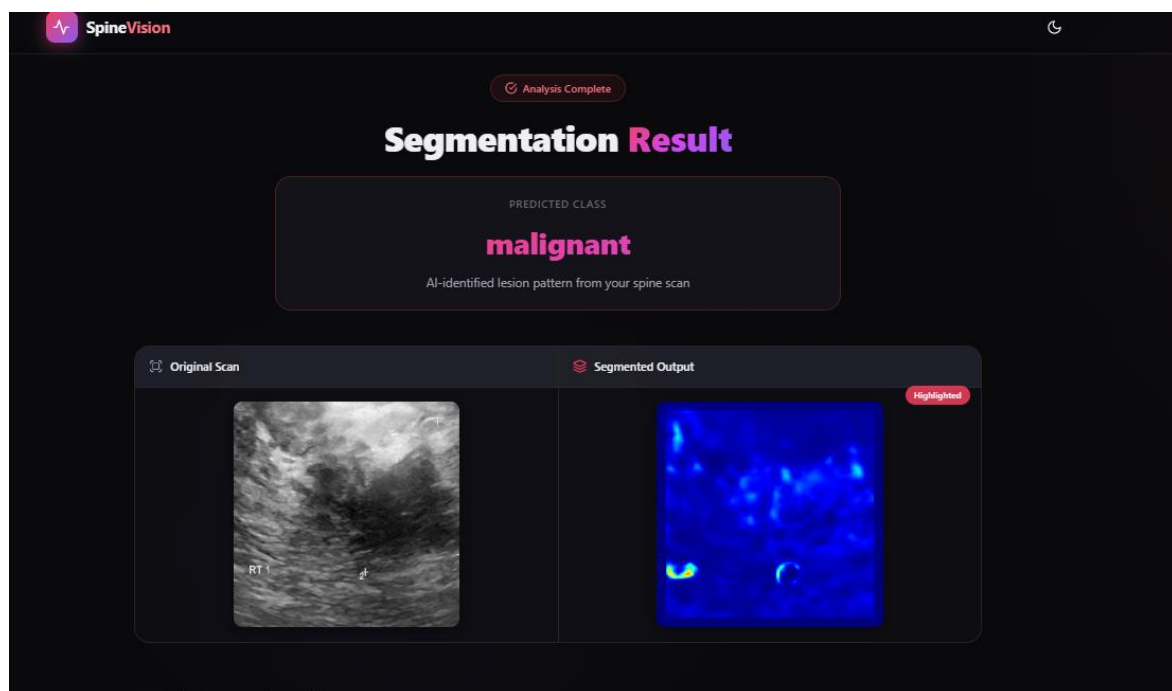
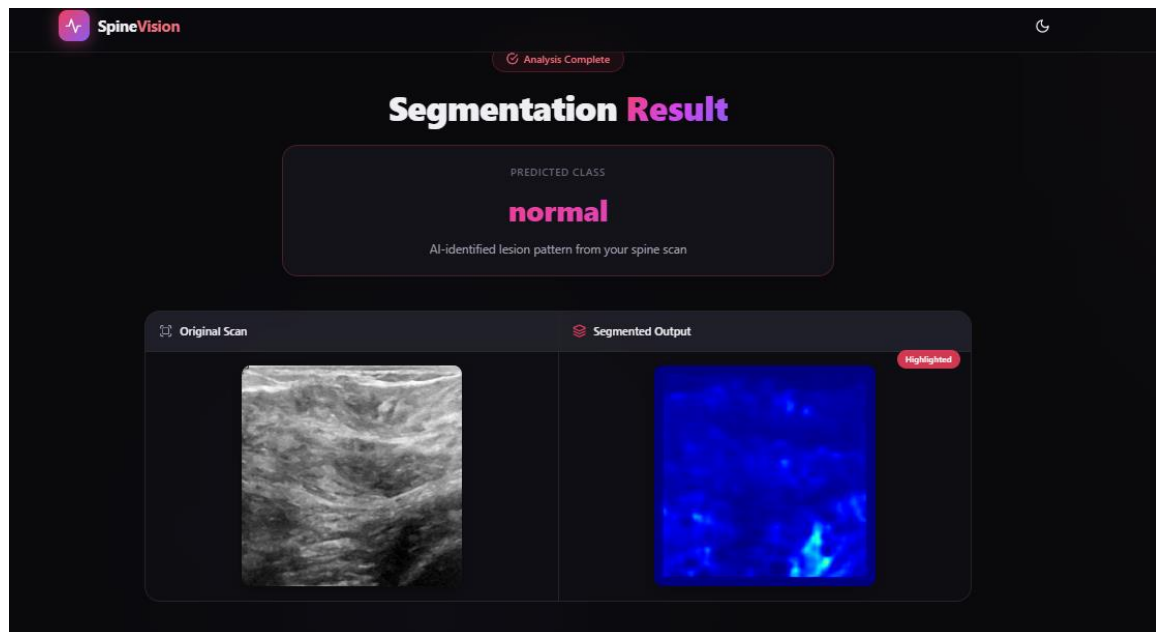
7.1 GENERAL:

This project is implemented like an application using Python and the Server process is maintained using the SOCKET & SERVERSOCKET and the Design part is played by Cascading Style Sheet.

7.2 SNAPSHOTS







CHAPTER 8

SOFTWARE TESTING

8.1 GENERAL

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

8.2 DEVELOPING METHODOLOGIES

The test process is initiated by developing a comprehensive plan to test the general functionality and special features on a variety of platform combinations. Strict quality control procedures are used. The process verifies that the application meets the requirements specified in the system requirements document and is bug free. The following are the considerations used to develop the framework from developing the testing methodologies.

8.3 TYPES OF TESTS

8.3.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program input produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

8.3.2 Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

- Valid Input : identified classes of valid input must be accepted.
- Invalid Input : identified classes of invalid input must be rejected.
- Functions : identified functions must be exercised.
- Output : identified classes of application outputs must be exercised.
- Systems/Procedures: interfacing systems or procedures must be invoked.

8.3.3 System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

8.3.4 Performance Test

The Performance test ensures that the output be produced within the time limits, and the time taken by the system for compiling, giving response to the users and request being send to the system for to retrieve the results.

8.3.5 Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

8.3.6 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Acceptance testing for Data Synchronization:

- The Acknowledgements will be received by the Sender Node after the Packets are received by the Destination Node
- The Route add operation is done only when there is a Route request in need
- The Status of Nodes information is done automatically in the Cache Updation process

8.2.7 Build the test plan

Any project can be divided into units that can be further performed for detailed processing. Then a testing strategy for each of this unit is carried out. Unit testing helps to identify the possible bugs in the individual component, so the component that has bugs can be identified and can be rectified from errors.

CHAPTER 9

FUTURE ENHANCEMENT

9.1 FUTURE ENHANCEMENTS:

Future works include the analysis of a multi-class classification method for this task, as well as an extension of the framework to MRI and radiography images.

CHAPTER 10

CONCLUSIONAND REFERENCES

10.1 CONCLUSION

In conclusion, the proposed system for the automated detection of spinal lesions from CT scans using the Xception model represents a significant improvement over existing methods. By leveraging deep transfer learning, the system eliminates the need for manual intervention, making the process faster and more efficient. The use of advanced preprocessing and the Xception model for feature extraction and classification enhances the accuracy and reliability of lesion detection, including a variety of lesion types such as metastases, tumors, and fractures. Additionally, the user-friendly web interface streamlines the diagnostic process, allowing healthcare professionals to review results quickly and make informed decisions. Overall, the proposed system offers a more scalable, accurate, and efficient solution for spinal lesion detection, providing valuable support in clinical decision-making and improving patient outcomes.

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Research Paper

AUTOMATED DETECTION OF SPINAL LESIONS FROM CT SCANS VIA DEEP TRANSFER LEARNING

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ABSTRACT— Automated detection of spinal lesions from Computed Topographies (CTs) can significantly enhance diagnostic efficiency and accuracy. This project implements a Computer-Aided Detection (CAdE) system for spinal lesion analysis using the Xception model. The system is equipped with an intuitive web-based interface developed using the Flask framework, allowing physicians to seamlessly integrate it into their diagnostic workflow. The CAdE system processes input CT scans by extracting vertebral regions of interest, converting them into 2D slices, and performing preprocessing to ensure optimal input quality for the Xception model. The model, fine-tuned with Transfer Learning, classifies vertebrae as either healthy

conditions such as metastases, primary tumors, or sclerotic lesions. The training and testing datasets were created from CT scans of patients Records. Data augmentation techniques were applied to expand the dataset and improve model generalization. The Xception model achieved a high accuracy of and a recall of 92.99%, demonstrating its effectiveness in spinal lesion detection. This system aims to provide a reliable and efficient tool to assist medical professionals in spinal lesion diagnosis, enhancing clinical decision-making and patient outcomes.

Index Terms – Spinal lesion detection, computed tomography (CT), Xception, transfer learning, deep learning, vertebra classification, Flask web application, diagnostic imaging.

I. INTRODUCTION

Computer-Aided Detection (CADE) techniques for the analysis of medical images have grown significantly in importance and sophistication. These tools are particularly critical for extensive exams, such as Computed Tomographies (CTs), which cover large sections of the patient's body and demand meticulous attention from medical professionals. Recently, there has been an increasing focus on automated diagnostic tools powered by Artificial Intelligence (AI). Among these, advanced deep learning models have proven highly effective for analyzing CT, MRI, ultrasound, and radiographic images, leading to state-of-the-art performance in various medical tasks, including spinal pathology detection and classification. Spinal metastases, a type of tumor that originates in other organs and spreads to the spine, are a significant diagnostic challenge. It is estimated that 50% to 70% of cancer patients develop bone metastases, with the spine being one of the most commonly affected regions. Diagnosing spinal tumors using 3D imaging such as CT or MRI is particularly challenging due to the intricate structure of the vertebrae and the sheer number of slices requiring analysis. This highlights the urgent need for a fully

automated detection system that can accurately and efficiently identify pathological features in the vertebral body. In this project, we present an automated CADE system utilizing the Xception model, a state-of-the-art deep learning architecture optimized for image classification tasks. Leveraging Transfer Learning, the Xception model is fine-tuned to detect abnormalities in vertebrae, including metastases, primary tumors, and other lesions. The system preprocesses CT scans, extracts vertebral slices, and analyzes them for potential anomalies. To enhance usability, the CADE system incorporates a web-based graphical interface developed using the Flask framework, allowing seamless integration into clinical workflows. This Xception-based solution demonstrates high diagnostic accuracy and provides a reliable, efficient tool to support medical professionals in spinal lesion detection, ultimately improving patient outcomes. Deep learning methodologies have shown promising results in cancer detection, segmentation, and classification, see, e.g., the recent survey.

Examples of the different cases of study to which deep learning has been applied include the detection, segmentation, and classification of breast cancer, lung cancer [5], brain tumors, skin lesion [8], liver tumor,

and musculoskeletal diseases. Studies about viral and bacterial pneumonia detection from X-rays were made. Latest research focuses on COVID-19 detection from thoracic CT scans [13], [14], [15]. These works consider different imaging problems, in which the inputs come from CT scans, X-Ray, MRI, or RGB images [8]. These methods use open-source networks like, e.g., UNet [16] and SegNet [17] suitably modified. When dealing with small datasets like in medical applications, training can be a difficult task. For this reason, one of the latest approaches is Transfer Learning, i.e. the practice of using state-of-the-art networks that have already been trained from huge datasets and then transferring the achieved knowledge to another domain. Inspired by these approaches, in this paper, we consider the development of a deep-learning based CADe system for spine lesions in CT scans. Focusing on the spine, a considerable number of studies regard the identification of the vertebral body volume, see, e.g., the work in [18]. In these works, the goal is to identify and label vertebrae, rather than classify eventual diseases. Several works focus on scoliosis and spondylolisthesis detection and classification, see the survey [19], and the recent works [20], [21], [22]. Differently from our work, these approaches do not focus

on vertebral lesions. Moreover, papers in [21] and [22] consider X-Ray images, and the work [20] considers ultrasound data. On the other hand, we consider CT scans. As for the segmentation of intervertebral disks, authors in [23] propose a harmonization approach based on GANs. Differently from our approach, they do not consider lesion classification. Moreover, their method is tailored for MRI images. Lumbar diseases are instead considered in [24]. Similarly to our method, authors propose a classification scheme based on VGG networks. However, they mainly focus on MRI images and do not consider vertebral lesions. A grading system for intervertebral disk degeneration is proposed in [25]. Similarly to the work in [24], authors do not consider vertebral lesions, and their approach is tailored for MRI images. As regards spine tumors, the attention is entirely on spine metastases. Several works consider deep learning models for the detection of abnormalities in the spinal column. In [26], authors propose a study of the performances of a Faster R-CNN network on MRI images. In [27] authors exploit Siamese networks to detect spine metastases from MRI images. Despite the high performance, the slices have to be manually extracted from the MRI sequences. In [28] MRIs are exploited to label vertebrae

and detect the presence of tumors. However, only 2D slices from the sagittal plane are considered, making small or non centered tumors hard to detect. As regards spine metastasis segmentation, [29], [30] use the UNet neural network on MRI images. Differently from these approaches, in this paper, we consider instead an automatic tool to detect spinal lesions from CT scans rather than MRIs. Also, the proposed methodology does not require manual annotations. As for deep learning methodologies for detection in CT scans, authors in [31] propose an unbalanced UNet for labeling and segmentation of vertebrae, without a particular focus on abnormalities. The work in [32] presents a pipeline based on a region of interest detector and a multi-modal classifier for spinal metastases. Unlike our approach, that study focuses on metastatic epidural spinal cord compression. In [33], authors propose a diagnosis method for benign and malignant vertebral fractures using the ResNet50 network. However, their pipeline relies on sagittal scans and requires human intervention to select the initial region of interest, which differs from our fully automated system. In [34], a deep learning-based tool employing a convolutional network is used for hybrid detection-segmentation of spinal metastases from CT

scans. Although effective, this system was tested exclusively on patients with high levels of metastases, without considering healthy controls. In contrast, our approach evaluates both healthy and lesioned vertebrae. The method in [35] focuses on segmenting metastases from manually extracted volumes of interest in CT scans. Our methodology, however, automates this process without requiring manual extraction. Authors in [36] propose a detection system specifically for lytic spinal lesions using a convolutional model, but their approach excludes other lesion types and does not incorporate a Computer Aided Detection (CADe) system for automated diagnosis. Similarly, [37] uses an iterative random forest-based method for detecting and classifying metastatic lesions in sagittal scans, which, while achieving good sensitivity, resulted in numerous false positives. For segmentation and detection of sclerotic metastases, the work in [38] utilizes a watershed algorithm on CT scans of the thoracolumbar spine

II. RELATED WORK

A. The use of deep learning in medical imaging to improve spine care: A scoping review of current literature and clinical applications

Medical imaging is critical for clinical decision-making and is integral to determining treatment indications and surgical outcomes. Various imaging modalities can be used for accurate detection and diagnosis of spinal pathologies [91]. However, despite following similar diagnostic standards, experienced radiologists can arrive at different diagnoses and measurements with error rates estimated to be 3% to 5% [7]. Much higher discrepancy rates are reported in neuroradiology, with variable reads in up to 21% of imaging studies [8]. Therefore, a key challenge in the diagnosis of spine pathology is improving the workflow to diagnose diseases quickly, automatically, and accurately.

Further, medical care costs are increasing [3], and radiologists who train for 5 or more years after medical school represent an expensive and valuable resource. AI tools to augment the performance of radiologists and provide a low-cost tool to prevent errors hold significant promises for improved quality and cost savings.

Reliable treatment planning, precise intervention, and accurate therapy are required when treating spine conditions to achieve optimal outcomes. While modern imaging techniques inform perioperative case management [9], poor outcomes are still

frequent. Thus, the development of scalable, perioperative automated assistance tools could help address this challenge.

B. Oncologic Applications of Artificial Intelligence and Deep Learning Methods in CT Spine Imaging

In spinal oncology, integrating deep learning with computed tomography (CT) imaging has shown promise in enhancing diagnostic accuracy, treatment planning, and patient outcomes. This systematic review synthesizes evidence on artificial intelligence (AI) applications in CT imaging for spinal tumors. A PRISMA-guided search identified 33 studies: 12 (36.4%) focused on detecting spinal malignancies, 11 (33.3%) on classification, 6 (18.2%) on prognostication, 3 (9.1%) on treatment planning, and 1 (3.0%) on both detection and classification. Of the classification studies, 7 (21.2%) used machine learning to distinguish between benign and malignant lesions, 3 (9.1%) evaluated tumor stage or grade, and 2 (6.1%) employed radiomics for biomarker classification. Prognostic studies included three (9.1%) that predicted complications such as pathological fractures and three (9.1%) that predicted treatment outcomes. AI's potential for improving workflow efficiency, aiding decision making, and reducing complications is discussed, along

with its limitations in generalizability, interpretability, and clinical integration. Future directions for AI in spinal oncology are also explored. In conclusion, while AI technologies in CT imaging are promising, further research is necessary to validate their clinical effectiveness and optimize their integration into routine practice.

C. Deep learning system for automated detection of posterior ligamentous complex injury in patients with thoracolumbar fracture on MRI

This study aimed to develop a deep learning (DL) algorithm for automated detection and localization of posterior ligamentous complex (PLC) injury in patients with acute thoracolumbar (TL) fracture on magnetic resonance imaging (MRI) and evaluate its diagnostic performance. In this retrospective multicenter study, using midline sagittal T2-weighted image with fracture ($\pm$ PLC injury), a training dataset and internal and external validation sets of 300, 100, and 100 patients, were constructed with equal numbers of injured and normal PLCs. The DL algorithm was developed through two steps (Attention U-net and Inception-ResNet-V2). We evaluate the diagnostic performance for PLC injury between the DL algorithm and radiologists with different levels of experience. The area under the curves

(AUCs) generated by the DL algorithm were 0.928, 0.916 for internal and external validations, and by two radiologists for observer performance test were 0.930, 0.830, respectively. Although no significant difference was found in diagnosing PLC injury between the DL algorithm and radiologists, the DL algorithm exhibited a trend of higher AUC than the radiology trainee. Notably, the radiology trainee's diagnostic performance significantly improved with DL algorithm assistance. Therefore, the DL algorithm exhibited high diagnostic performance in detecting PLC injuries in acute TL fractures.

D. Deep learning-based automated spine fracture type identification with Clinically validated GAN generated CT images

Automatic type identification of sub-axial spine fractures is of prime importance for orthopedic to reduce image interpretation time and increase patient care time. But identifying fracture types is challenging due to imbalanced datasets. In this work, CT scan images of fractured spine has been collected from a Tertiary Care hospital and extended Deep Convolutional Generative Adversarial Network (DCGAN) architecture is developed for generating spine fracture images that overcomes the imbalanced dataset problem.

These enhanced datasets are clinically evaluated with Two Visual Turing Tests (VTTs): the first test to “identify real and generated images” and second test to determine “type of fractures in the generated images.” The first VTT demonstrates that generated images of fractures are realistic and that even spine surgeons have difficulty in distinguishing them from real. The second VTT demonstrates that fracture lines are clearly visible in the generated images. The VTT results are analyzed using Fleiss Kappa statistical techniques to determine the inter observer reliability of spine surgeons’ clinical evaluation. The results showed high interobserver agreement for “type identification” in the generated images. The clinically evaluated generated images are provided to the proposed ensemble-based type identification model, which outperformed other models in type identification.

III. PROPOSED METHODOLOGY

The proposed system uses the advanced Xception model, which leverages separable convolutions to improve feature extraction and overall performance. It eliminates the need for manual region selection and can automatically detect and classify a wide range of spinal lesions, including metastases, tumors, and fractures. The system also

includes a user-friendly interface built with Flask, allowing medical professionals to easily upload CT scans and receive diagnostic results. Furthermore, the proposed system incorporates advanced preprocessing techniques, such as data augmentation, to enhance its accuracy and adaptability to varying medical imaging data. Overall, the proposed system offers a more automated, accurate, and scalable solution for spinal lesion detection compared to existing methods.

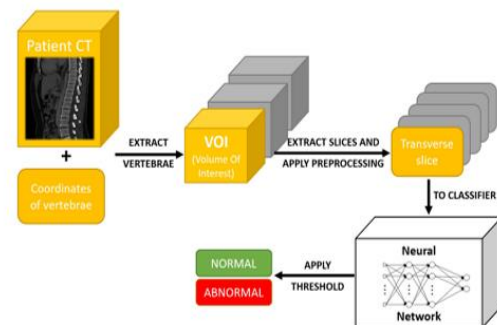


Fig. 1: System Overview

1. Data Collection

This step involves gathering ultrasound image data related to histotripsy ablation. The data could include images of both treated (ablated) and untreated (non-ablated) tissue, usually acquired from patients undergoing the histotripsy procedure or from publicly available datasets.

2. Data Analysis/Load Data

After collecting the dataset, the next step is analyzing and loading the data into the system. This involves checking the quality of the data, ensuring there are no missing or corrupted images, and formatting them correctly for further processing. In this step, the images are typically resized and normalized for consistency.

3. Data Feature Extraction

In this step, the system uses various techniques to extract meaningful features from the ultrasound images. Features such as texture, edges, or regions with high intensity (indicating potential ablation) are identified. This helps the Xception in the next step to learn essential patterns from the images.

4. U-Net Model for Segmentation

The U-Net model is a deep learning architecture specifically designed for image segmentation. It consists of an encoder (which captures feature representations of the image) and a decoder (which reconstructs the segmentation output). The U-Net is used to segment the ultrasound images into regions of ablated and non-ablated tissues, providing a pixel-wise classification.

5. Split Dataset

To train and evaluate the model, the dataset is divided into three parts: training,

validation, and testing datasets. Typically, the dataset is split such that 60% is used for training the model, 20% for validating the model during training (to tune hyperparameters), and the remaining 20% for testing the final model's performance.

6. Xception Model for Training

In this step, the Xception model is trained using the segmented images from the U-Net model. The Xception is responsible for learning high-level features from the images and classifying them as ablated or non-ablated tissues. The model uses the images from the training dataset to learn patterns that are then applied to unseen test data.

7. Model Evaluation

After training the Xception, the model is evaluated using the test dataset. Metrics such as accuracy, precision, recall, and the Dice similarity coefficient are used to assess how well the model performs. This step helps in determining whether the model has learned to correctly identify ablation areas and if further tuning or adjustments are needed.

8. Model Save

Once the model achieves satisfactory performance, it is saved for future use. The trained model can be saved as a file (e.g., .h5 or .pkl) and later loaded to make predictions

on new, unseen ultrasound images. Saving the model ensures that it does not have to be retrained every time it needs to be used in a real-time application.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The experimental results show that the proposed Xception-based deep transfer learning model accurately detects and classifies spinal lesions from CT scans. The use of transfer learning and data augmentation improves model generalization and diagnostic performance, achieving a recall of 92.99% with high classification accuracy. The integrated Flask-based CADe system enables efficient and user-friendly clinical diagnosis.

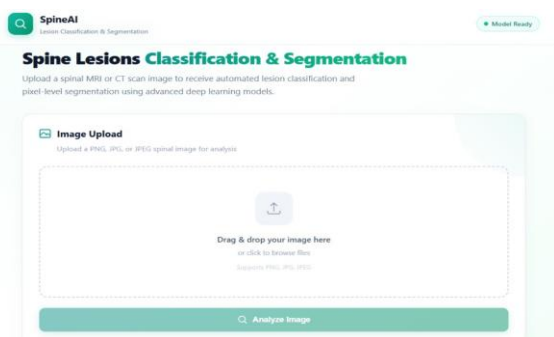


Fig. 2: Upload Image

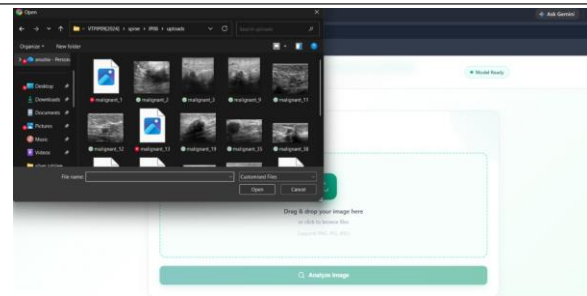


Fig. 3: Choose Image

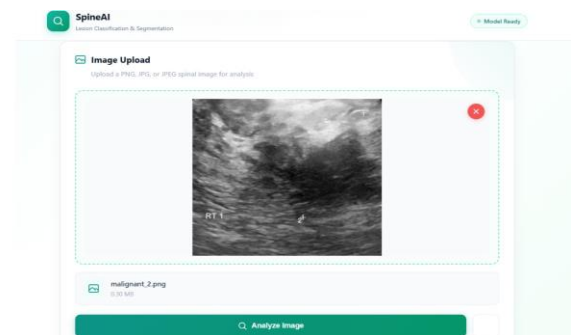


Fig. 4: Analyze Image

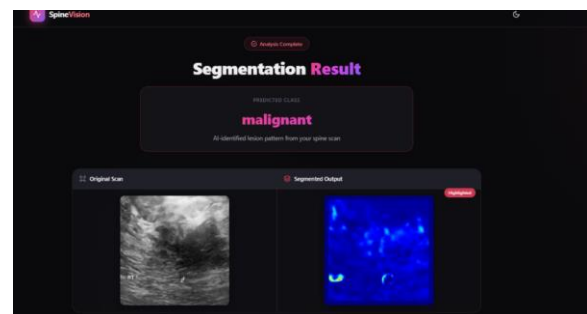


Fig. 5: Results

V. CONCLUSION

In conclusion, the proposed system for the automated detection of spinal lesions from CT scans using the Xception model represents a significant improvement over existing methods. By leveraging deep transfer learning, the system eliminates the need for manual intervention, making the

process faster and more efficient. The use of advanced preprocessing and the Xception model for feature extraction and classification enhances the accuracy and reliability of lesion detection, including a variety of lesion types such as metastases, tumors, and fractures. Additionally, the user-friendly web interface streamlines the diagnostic process, allowing healthcare professionals to review results quickly and make informed decisions. Overall, the proposed system offers a more scalable, accurate, and efficient solution for spinal lesion detection, providing valuable support in clinical decision-making and improving patient outcomes.

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